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Next Generation Water Prediction Capability (NGWPC) EDFS Missing Reservoir Documentation

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1.0 INTRODUCTION

1.1 Purpose

This document describes the development of the reservoirs, lakes, and dams datasets included in the NGWPC Hydrofabric (NHF). It identifies the upstream source datasets, explains the selection criteria used to determine which locations were included, and describes how those sources were used to build the lakes, lakes_polygons, lake_vfp_crosswalk, and reservoir_da layers.

2.0 NHF Lake Architecture

2.1 Lakes – Points

The “lakes” layer is a point layer encompassing all lakes, dams, and reservoirs in the NHF. The lakes table is used directly by T-Route. It includes all attributes necessary for T-Route, USACE NID ID where available, run-of-river flag, and the source dataset of the lake.

2.2 Lakes – Polygons

The “lakes_polygons” layer is a polygon layer. It provides the polygon representation of all lakes in the “lakes” point layer. Polygons are derived from the sources described below in section 3.0: NWM v3 lakes, reference waterbodies, and manually generated in instances in which there was not a previously derived polygon representation (e.g. run-of-river and low-head dams).

2.3 Lake – Virtual Flowpath Crosswalk

The “lake_vfp_crosswalk” layer is a non-spatial table that intersects virtual flowpaths with lake polygons. This creates a many:many relationship where a lake may have one or more intersecting virtual flowpaths and a virtual flowpath may intersect one or more lakes. The case of a virtual flowpath intersecting multiple lakes is rare (1.2% of virtual flowpaths intersecting lakes) but possible because the underlying reference fabric creating the network is not segmented based on lake outlets. This crosswalk is used in T-Route to assign the correct routing to virtual flowpaths within a lake.

2.4 Reservoir DA

The “reservoir_da” layer is a non-spatial table that supports the reservoir data assimilation (DA) methods in T-Route. It removes the need for T-Route to calculate these complex reservoir:gage relationships. The table crosswalks lakes to stream gages where possible and assigns a DA scheme number to all lakes.

3.0 NHF Lake Geometry Sources

The following sources are used to create *geometries* for lakes.

3.1 NWM v3 Lakes

OWP provided a polygon layer of all NWM v3 lakes with NHD 2.1 COMID (labeled as “newId” in the layer). All polygons are retained in NHF as both points (“lakes”) and polygons (“lakes_polygons”). There are 6,044 lakes that have been included across all domains.

3.2 Reference Waterbodies

Reference Waterbodies is a polygon layer linked to NHD 2.1 COMID. It includes additional waterbodies that do not exist in NWM lakes. Geometries for a given COMID may not be the same as NWM v3 lakes. Reference waterbodies are only included when reservoir DA sources are not present in NWM v3 lakes.

3.3 Reference Reservoirs

Reference Reservoirs is a point dataset fusing multiple reservoir datasets together to achieve the best possible point location of dams. Datasets included were the National Inventory of Dams (NID), ResOpsUS, Global Dam Watch (GDW), Hydropower Infrastructure – Lakes, Reservoirs, Rivers (HILLARI), and Open Street Map (OSM). As you recall, this reference reservoirs dataset was delivered in PI-7.

Data from the USACE NID was used to supplement the dams with structural information. Reference Reservoirs were used to enhance the point geometry of NWM v3 and Reference Waterbodies by moving the point location to the outlet of the lake when possible.

Reference Reservoirs were also added to the lakes dataset if they met the following criteria: crosswalked to a polygon (NWM lakes v3 or Reference Waterbodies), minimum 0.2 km² surface area, and a maximum of 1 km from a flowpath. These settings were designed to be configurable in the NHF lakes > ref_res settings for `min_wb_area_sqkm` and `max_distance_m`.

3.4 Manual Polygons

Manual polygons were created for the large RFC run-of-river dams (Section 4.1.3) and low-head dams (Section 4.3.4) where no polygon was available from existing datasets. The remaining Section 4.0 datasets are crosswalked to an existing lake in NWM v3 Lakes (Section 2.1) or Reference Waterbodies (Section 2.2). Supplemental RFC gages added during this effort are retained as source data to support future crosswalking and integration.

4.0 Reservoir DA Sources

4.1 RFC

River Forecasting Center (RFC) locations were obtained from the sources listed below.

4.1.1 NWM v3 RFC

NWM v3's "reservoir_index_AnA.netcdf" includes a crosswalk between lake_id/COMID and gages. 397 RFC gages were extracted and added to the NHF. A total of 299 were marked as RFC-DA and 102 sites were found to be inactive when referenced in the National Water Prediction Service (NWPS) spreadsheet "all_gauges_report.csv" that can be found on NOAA's NWPS data downloads site

(https://water.noaa.gov/resources/downloads/reports/nwps_all_gauges_report.csv).

Inactive gages have been set to level pool.

4.1.2 Ohio RFC

OWP provided a list of Ohio River Valley RFC locations. These RFC points were manually crosswalked to corresponding lake polygons using manual visual inspection. 76 RFC's were added and set to RFC-DA scheme. There were 4 RFC's that could not be crosswalked as there was no known lake polygon available. Consequently, these 4 sites have been excluded from the current NHF until more information is available.

4.1.3 Large Run-of-river

Large run-of-river RFC dams were identified using Google AI, per OWP direction. Dams were joined to USACE NID attributes. Out of 12 identified dams with RFC codes, 2 were crosswalked to existing lake polygons and 10 were represented with a new polygon.

4.1.4 Additional RFC

We were able to acquire 2,697 additional gages from the NWPS dataset "all_gauges_report.csv"

(https://water.noaa.gov/resources/downloads/reports/nwps_all_gauges_report.csv) available through NOAA's NWPS data downloads page and added them to the dataset. The required acquisition of gage data for this delivery is complete, and these supplemental records provide additional source data that can support future crosswalking and integration work.

All gages used in NWM v3's "reservoir_index_AnA.netcdf" (section 4.1.1) were retained, regardless of status.

4.2 USGS

NWM v3's "reservoir_index_AnA.netcdf" includes a crosswalk between lake_id/COMID and gages. 76 USGS gages were marked as USGS DA in the reservoir_da table.

4.3 USACE

The United States Army Corps of Engineers (USACE) provides dam data from multiple sources. The following sources were integrated into NHF.

4.3.1 National Inventory of Dams

The USACE NID was used to enhance lake attributes where available. NID was used as a source in the reference waterbodies, large run-of-river dams, and low-head dam inventory. These structural attributes are used in the hydraulic parameter generation to produce more accurate estimates of T-Route parameters.

4.3.2 NMW v3 USACE

NWM v3's "reservoir_index_AnA.netcdf" includes a crosswalk between lake_id/COMID and gages. 70 USACE gages were crosswalked and marked as USACE DA in the NHF reservoir_da table.

4.3.3 Additional USACE

The USACE Water Management System Data (CWMS) API was queried to add additional USACE gage/dam locations where flow data was available. First, the location endpoint was queried with `curl https://cwms-data.usace.army.mil/cwms-data/location/group\?location-category-like\=flow` to find USACE locations that included flow data. Then, the location and office ID were queried to retrieve latitude and longitude of the site. An example query is `curl https://cwms-data.usace.army.mil/cwms-data/locations/BIGH\?office=SWT` which returns the site BIGH (Big Hill Lake) and its geometry. All sites in the flow category with geometry were downloaded. Sites were crosswalked to lakes within NWM lakes and reference waterbodies where possible using the `geopandas.sjoin_nearest` function with a max buffer of 300 meters. There were 177 sites that were added and marked as USACE DA in the reservoir_da table.

4.3.4 Low-head Dam Inventory

The USACE Low-head Dam Inventory includes confirmed and probable low-head dams. In all, 210 confirmed dams were added to NHF. These low-head dams were crosswalked to lake polygons using the `geopandas.sjoin_nearest` command with a buffer of 300 meters. 21 of the dams were able to be crosswalked to existing polygons. The remaining 189 were given a representative buffer polygon around the point. Since low-head dams will be modeled as channels, the lake parameters did not require high precision. Low-head dams were flagged with the 'run_of_river' flag in the lakes table and are currently modeled as channel in T-Route.

4.4 USBR

USBR sites were acquired from the USBR RISE ArcGIS Feature Service. The feature service was queried for any site listed as "Lake/Reservoir". The sites were then crosswalked to lake polygons using spatial buffers and finally individually manually

inspected for accuracy. USBR lakes were set to USBR reservoir DA scheme in the reservoir DA table. In total, 260 USBR sites were added to the NHF.

4.5 NRCS

NRCS dams were tagged in NHF if the owner of the dam included the string "NRCS" for lakes with NID data. The `source` column concatenates the original source of the lake (e.g. NWM, low-head dams) with `\_NRCS` to indicate the joint source of the dam data. Lakes have been flagged as NRCS.

5.0 Acronyms, Abbreviations, and Notes

Acronym	Description
API	Application Programming Interface
AWS	Amazon Web Services
AWS Glue	AWS serverless data integration service
BIGH	Big Hill Lake
COMID	Common Identifier
CONUS	Continental United States
CPU	Central Processing Unit
CWMS	Corps Water Management System
DA	Data Assimilation
DOI	Department of Interior
GDW	Global Dam Watch
HILLARI	Hydropower Infrastructure – Lakes, Reservoirs, Rivers
NHF	NGWPC Hydrofabric
NID	National Inventory of Dams
NOAA	National Oceanic and Atmospheric Administration
NRCS	Natural Resources Conservation Service
NWM	National Water Model
NWPS	National Water Prediction Service
OWP	Office of Water Prediction
OSM	Open Street Map
PET	Positron Emission Tomography
RISE	Reclamation Information Sharing Environment – USBR's centralized data portal
RFC	River Forecast Center
SAC-SMA	Sacramento Soil Moisture Accounting
SNOW-17	Snow accumulation model
SuperCONUS	Continental US + adjacent Mexico and Canada drainages
SQL	Structured Query Language
TWI	Two-Way Immersion
USACE	US Army Corps of Engineers
USBR	US Bureau of Reclamation
USGS	US Geological Service